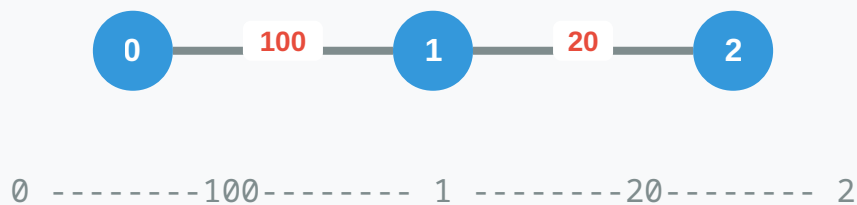


Chapter: Graphs & Shortest Paths

Concept & Highway Tolls

Each highway has a toll.

Example:



If you travel

0 → 1 → 2

you pay

100 + 20 = 120

The Twist: Using Discounts

Now comes the twist

You are given **discounts = 1**

A discount can be used on any one highway.

When you use it,

You pay $\text{floor}(\text{toll} / 2)$

Example

Highway cost

100

Using discount

$100/2 = 50$

Another example

Highway cost

9

Using discount

$\text{floor}(9/2) = 4$

State Representation & Initialization

we'll use

$\text{dist}[\text{node}][\text{discountRem}]$

Meaning

Minimum cost to reach this node while still having discountRem discounts left.

Initially

Suppose

$\text{discounts} = 3$

Initially

$\text{pq.push}(\{0, \{\text{src}, 3\}\});$

because we haven't used any discounts.

Also

```
dist[src][3]=0;
```

Transitions / Choices

Choice 1

Don't use discount

Pay

80

State

```
newdiscountRem =  
currdiscountRem;
```

Nothing changes.

Choice 2

Use discount

Pay

40

State

```
newdiscountRem =  
currdiscountRem-1;
```

C++ Code Implementation

```

class Solution {
public:
    int minimumCost(int n, vector<vector<int>>& highways, int
discounts) {

        vector<vector<pair<int,int>>> adj(n);
        for(auto &it:highways){
            int u=it[0];
            int v=it[1];
            int toll=it[2];
            adj[u].push_back({v,toll});
            adj[v].push_back({u,toll});
        }

        //minheap to store {cost,{node,discountremaining}}
        priority_queue<
        pair<long long,pair<int,int>>,
        vector<pair<long long,pair<int,int>>>,
        greater<pair<long long,pair<int,int>>>> pq;

        //dist[node][discountRemaining] = minimum cost to reach this
node
        //while still having 'discountremaining' discounts left
        vector<vector<long long>> dist(n,vector<long
long>(discounts+1,LLONG_MAX));

        pq.push({0,{0,discnts}});
        dist[0][discounts]=0;

        while(!pq.empty()){

            long long currdiss=pq.top().first;
            int currnode=pq.top().second.first;
            int currdiscountrem=pq.top().second.second;

            pq.pop();

            if(currdiss>dist[currnode][currdiscountrem]){
                continue;
            }

            if(currnode==n-1){
                return currdiss;
            }

            for(auto &it:adj[currnode]){
                int newnode=it.first;
                int toll=it.second;

                //Case 1 : bina discount use kiye

```

```

        long long newdiss=currdiss+toll;
        if(newdiss<dist[newnode][currdiscountrem]){
            dist[newnode][currdiscountrem]=newdiss;
            pq.push({newdiss,{newnode,currdiscountrem}});
        }

        //ek discount use kr lo
        if(currdiscountrem>0){
            newdiss=currdiss+(toll/2);
            if(newdiss<dist[newnode][currdiscountrem-1]){
                dist[newnode][currdiscountrem-1]=newdiss;
                pq.push({newdiss,{newnode,currdiscountrem-1}});
            }
        }
    }
    return -1;
}
};

```